Predicting Game Genres From Cover Arts Using CNNs

Motivation & Problem Statement

- Video game covers are designed to instantly convey genre, mood, and style
- A model that predicts video game genres according to covers can be used to improve game discoveries, recommendations, and archives
- Since a game can fall under multiple genres, this is a multi-label classification problem
- Baseline model GameNet-1 achieves ~30% accuracy on a simpler single-label task
- Goal: Build a custom CNN that outperforms GameNet-1 on the harder multi-label task applied to 33 Steam genres

Dataset

- Utilized the Steam Games 2025 Kaggle dataset
- After removing games with no genre tags, we were left with 89,403 game covers spanning 33 genres (Action, Adventure, Indie, RPG, etc.)
- All images were resized to 224x224 and normalized using ImageNet mean and std
- Split: 70% training / 30% test
- Note: There is a severe class imbalance - Indie, Action, Adventure, and Casual dominate, rarer genres only had a few samples

GameNet-1 Model

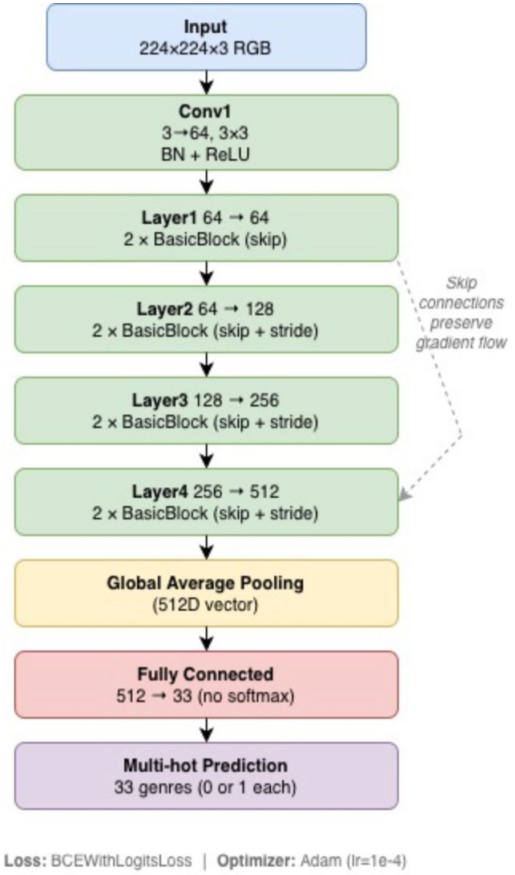
- GameNet-1 is a CNN model that focuses on trying to predict the genre(s) for popular video games based on their cover art, with it using a pretrained EfficientNetB3 model that was finetuned for this task as the base for this model

- This model was trained using the original Steam Games Dataset, with the training occurring in 2 phases along with using the Adam optimizer and categorical cross-entropy loss for training the model

- Even though this model wasn't trained with the Steam Games 2025 dataset like what we used for our model, the overall data format and how the data is being represented is retained between the 2 datasets, making it a viable baseline

Methodology

- Built a custom ResNet-18 in PyTorch (no pretrained weights)
- Multi-label output, final layer has 33 logits
- Utilized Binary Cross-Entropy with Logits Loss function
- Utilized Adam optimizer with a learning rate of .0001
- Trained with a batch size of 8 and 10 epochs
- Model was saved whenever validation loss improved
- Performance was compared directly with GameNet-1 baseline



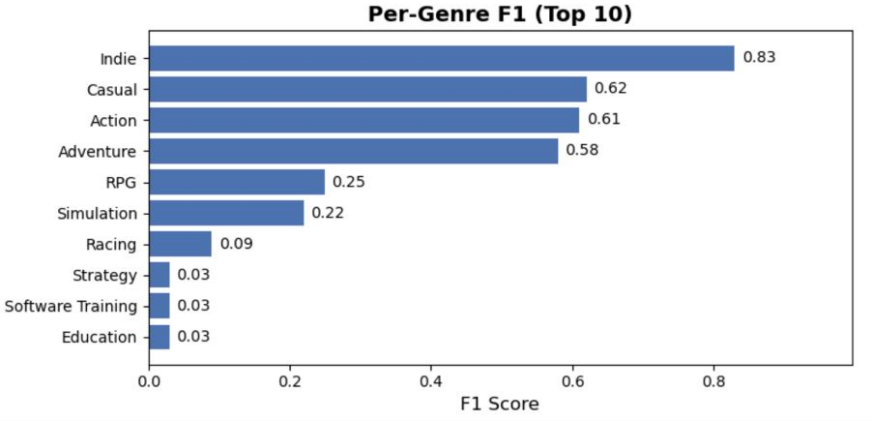
Evaluation

- Test set contained 26,821 unseen covers (30% of total covers)
- Micro F1: 0.578 - main multi-label metric, almost doubling GameNet-1's ~0.30
- Macro F1: 0.100 - model favors frequent genres
- Top performing genres:
  - Indie (0.83)
  - Casual (0.62)
  - Action (0.61)
  - Adventure (0.58)
- Our model outperforms the baseline on this harder multi-label task

Metric	Value
Element-wise Accuracy	0.928
Micro F1	0.578
Macro F1	0.100
Indie F1	0.83
Casual F1	0.62
Action F1	0.61
Adventure F1	0.58

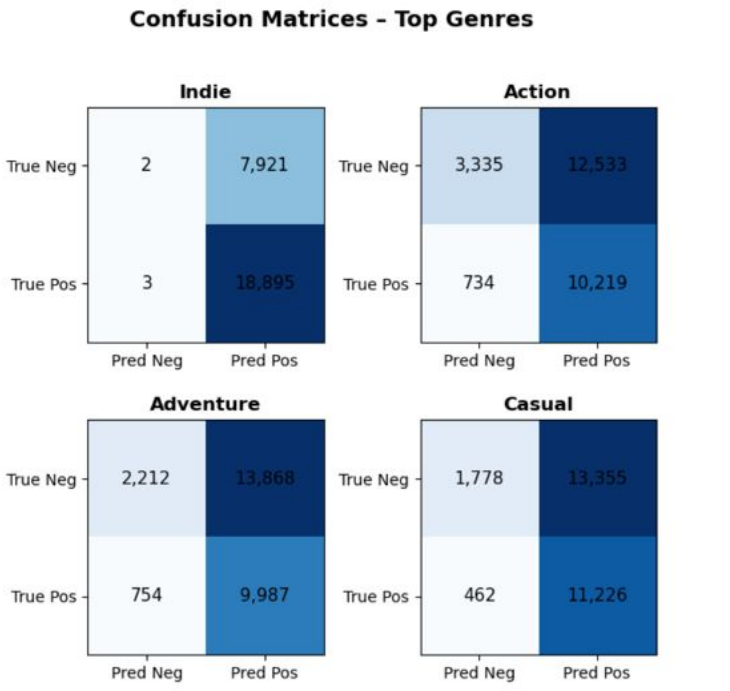
Per-Genre Performance

- The four main genres break the 0.5 F1 barrier and dominate predictions
- Rare genres (RPG, Strategy, Racing) have very low F1 scores
- Further work could focus on how the model addresses class imbalances



Confusion Matrices

- Indie: lots of true positives (TP), moderate amount of false positives (FP)
- Action/Adventure: similar pattern - high recalls, but lots of FP
- Casual: high recall, lots of FP
- Rare genres mostly yielded false negatives



Baseline Comparison

- GameNet-1: ~30% accuracy on a single-label task with a 5-genre subset
- Our model: Micro F1 of 0.578 on 33-genre multi-label task
- Direct comparison is difficult due to difference of tasks
- Our model handles a realistic, multi-label problem and significantly exceeds the baseline performance on a broader range of genres
- Our custom ResNet-18 is a clear improvement over GameNet-1

Discussion & Future Work

- Class imbalance results in over-prediction of common genres and ignores rare genres
- Future improvements:
  - Focal loss of weighted BCE to penalize easy negatives
  - Data augmentation / oversampling for rare genres
  - Decision threshold tuning per genre
  - Pretrained backbone
  - Hierarchical genre grouping to reduce label space
- Despite the class imbalance, the model delivers strong results on common genres, outperforming the baseline model

References

[1] GameNet-1 Model, https://huggingface.co/MAS-AI-0000/GameNet-1

[2] Steam Games Dataset, https://www.kaggle.com/datasets/fronkongames/steam-games-dataset

[3] Steam Games Dataset 2025,
https://www.kaggle.com/datasets/artermiloff/steam-games-dataset

[4] He, K., Zhang X., Ren, S., & Sun, J. Deep Residual Learning for Image Recognition. CVPR 2016,

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